NKG East Africa

STOCKTAKE OPTIMIZATION PROJECT

Business Requirements Document (BRD)

Volume 1 — Kenya Engagement

Item	Detail
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Prepared by	Pathways Technologies
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Business owner	Carol Muthama — Country Finance Manager, IBKY

Document Control

Version History

Version	Date	Author	Summary of changes
V1.0	October 2026	Pathways Technologies	Initial draft from the discovery session notes.
V1.1	9 October 2026	Pathways Technologies	Full review against the four session recordings: glossary, month-end calendar, reconciliation levels, unit rules, stock categories, consolidated requirements, exception catalogue, reports, risks, decisions and success measures.
V1.2	12 October 2026	Pathways Technologies	V1.0 and V1.1 consolidated into one client document. Wording simplified; all findings, pain points, proposed solutions, requirements and actions retained. Added Annex A — Solution Build Register.

Review and Approval

Role	Name	Signature	Date
Business Owner (approver)	Carol Muthama — Country Finance Manager		
Finance reviewer	Caroline Nzisa / Lucy Maundu		
Warehouse & Operations reviewer	Margaret Mburu / Luis Brian		
Trade & QC reviewer	Njambi Gatheru / Daniel Magu / Judy Murimi / Jad Belmezouar / Billy Rotich		
Logistics reviewer	Whitney Mwangi / Francis Magomba		
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How to Read This Document

- Sections 1–4 summarise the engagement, the shared vocabulary and the end-to-end stock cycle.
- Sections 5–8 describe each department in the same four parts: Role, As-Is Process, Reconciling Items Observed, and Pain Points and Proposed Solutions. Each pain point has an ID (F = Finance, W = Warehouse & Operations, T = Trade & QC, L = Logistics).
- Sections 9–12 consolidate the requirements (FR), exceptions (EX), reports and non-functional requirements (NFR).
- Sections 13–17 cover scope boundaries, risks, decisions needed, actions and success measures.

- Annex A is the Solution Build Register, completed by the delivery team to show what will be built and which pain point each item resolves.

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1. Executive Summary

1.1 Purpose

This document records the business requirements gathered during the discovery and requirements-gathering sessions held with the Finance, Warehouse & Operations, Trade & Quality Control and Logistics teams of Ibero Kenya Limited (IBKY) between 5 and 7 October 2026. It forms part of the Stocktake Optimization Project under the EA Controlling & Shared Services Digital Transformation programme, and follows the structure and standards of the master NKG Business Requirements Gathering Documentation and Analysis document.

It is the reference baseline for the design, build, testing and acceptance of the enhanced Stock Management solution across Power Apps, Power Automate, Power BI, the Azure Data Warehouse (governed data layer) and the Copilot Studio assistant. Every requirement carries an ID (Section 9) so it can be traced into design, test scripts and user acceptance.

1.2 Business Context

The project builds on work already delivered under the EA Controlling & Shared Services programme. The Stock Management Power App is in use at Ibero and the third-party warehouses, and every count is captured with counter identity and a timestamp, with each count line compared to XBS. The application stops at the point of capture: it does not reconcile physical counts against XBS balances, does not triangulate against Comtras/BRIC at basket level, and does not flag, classify or explain variances.

The April 2026 (P04) reconciliation took **33 calendar days** from period end (30 April) to final report circulation (2 June), against an SOP target of sign-off within **7 working days**. It generated **nine or more workbooks, 76 worksheets and about one million cells** of manual analysis, and **15–20 person-days** of forensic Excel work. The same error families recur every cycle, which shows the problem is structural rather than a one-off.

Finance confirmed that the first-level check (physical vs XBS) takes under an hour once the count is clean, but the second-level reconciliation against BRIC/Comtras takes about two weeks of manual tracing. Finance set a goal of removing roughly 20 days from the cycle.

The need is growing. Finance expects IBKY to roughly **double the inventory it handles**, with **direct purchases** playing a much larger role. Direct purchases are the hardest stock to track today, because contracted, warranted and delivered quantities differ and are followed in Excel.

The sessions confirmed that the next phase of value lies in extending the application into a full **reconciliation, exception-management and visibility platform** — building on what is already deployed, not replacing it.

1.3 Scope of This Engagement

Department	Focus areas
Finance	Month-end cut-off and close; physical vs XBS (Level 1) and XBS vs Comtras/BRIC (Level 2) reconciliation; variance analysis and reconciling items; unit conversion; direct-purchase tracking; adjustments and data corrections; exception management; sign-off; audit readiness; AI-assisted explanation

Department	Focus areas
Warehouse & Operations	Physical counting; stock categories in XBS (current, processing, WIP, dispatch); XBS data dependency and extract timing; receiving and processing; location management; third-party warehouse reconciliation; duplicate and multi-location lots; wrong-warehouse finds; loose bags, ex-office and sweepings; lights and pocket handling; live count dashboard; exception handling
Trade & Quality Control	Auction and direct purchasing; processing and basket classification incl. the new Low Grade basket and monthly downgrade policy; dispatch and lot blocking; allocation and stuffing; stuffed-not-invoiced; contract balances (XBS vs Comtras) and roll-over at 100 amendments; contract timing; three positions; provisional stock; franchise, ex-office, loose bags and sweepings; duplicate offers
Logistics	Movement of coffee into and out of Ibero warehouses; pre-count third-party confirmation; partial collections and part-lot tracking; warrant-location mismatches; post-count variance explanation; third-party ageing; batched data corrections; comment log

1.4 Key Findings

- **Three positions, no shared key.** Stock is recorded at three levels of detail — bags per stack (Power App), outturn/lot (XBS) and primary/secondary basket (Comtras/BRIC). The link between them is maintained by Trade during the month and rebuilt by Finance at month-end.
- **Level 2 is the bottleneck.** Physical vs XBS is quick once the count is clean; XBS vs Comtras takes about two weeks because of processing, downgrades, stuffed-not-invoiced coffee, rolled-over contracts and items that exist in only one system.
- **Discovery happens at month-end.** Unposted entries, duplicates, outturn mismatches and wrong warehouse allocations surface during or after the count rather than on the day they occur.
- **Timing, not loss, drives most variance.** Examples: direct sales after the count, data corrections and credit notes not yet posted, coffee awaiting another outturn, third-party stock delivered after the count, coffee stuffed but not invoiced.
- **Variance without explanation.** The dashboard shows what the variance is but not why. Explanations live in calls and emails, so they cannot be reused next month or by an AI assistant.
- **Incomplete inventory visibility.** WIP, processing, provisional, direct-purchase, sweepings, ex-office, loose bags, disputed and franchise stock are not consistently visible. Several are physically present but invisible to the app and XBS, and are reintegrated as "symbolic" direct purchases.
- **No period-end snapshot.** XBS downloads for the same date return different balances after processing, so teams do not work from a single version of the truth.
- **Baskets move after purchase.** Processing does not upgrade coffee; it separates lower-quality material. A fourth primary basket, Low Grade, was created and Trade plans to move processed downgrades monthly. This is not yet reflected consistently in XBS and Comtras.
- **Contract complexity.** Rolled-over contracts, residual balances and the timing gap between purchase, warrant and delivery (e.g. 1,000 → 900 → 800 bags) create structural variance that cannot be traced without manual effort.
- **Stuffed-not-invoiced variance.** Coffee stuffed but not yet invoiced creates a long/short difference between XBS and Comtras that must be aged and explained every cycle.

- **Weak traceability.** Stock movements cannot easily be traced by warehouse, zone, grid, location, lot and grade, which makes investigations and audits slow.
- **Weak third-party visibility.** Only two of five third-party warehouses return balances before the count, and stock is often found in a different warehouse from the warrant (e.g. Maxa instead of AGL).
- **Manual, paper and email controls.** Allocation reports are reformatted by hand, roll-overs are coordinated by email, third-party reports arrive in different formats, WhatsApp is used where a governed workflow should exist, and sign-off is printed and circulated.

1.5 Priority Requirements at a Glance

Ref.	Requirement
FR-01 to FR-05	Controlled XBS and Comtras/BRIC extracts at an agreed cut-off, approved and stored as a locked month-end snapshot
FR-10 to FR-14	Master mapping (outturn → quality code → secondary → primary basket → contract) and one set of conversion rules
FR-20 to FR-23	All stock categories tracked: provisional, franchise, ex-office, sweepings, loose bags, disputed, WIP and processing
FR-30 to FR-34	Count-day controls and live dashboard; duplicate, multi-location and wrong-warehouse capture; location management
FR-40 to FR-44	Two-level reconciliation (physical vs XBS; XBS vs Comtras/BRIC) with reconciling items and stuffed-not-invoiced ageing
FR-50 to FR-56	Exception register with root causes, owners, comments, alerts and batched corrections
FR-70 / FR-71	Electronic sign-off with timestamps and automatic distribution

1.6 Decisions Required from IBKY

Ten decisions are needed before design sign-off; they are listed with owners in Section 15. The most important are the official month-end cut-off and extract time, the unit and pocket rule shared by the app and XBS, how the Low Grade downgrade will be posted, and where provisional stock will be held.

2. Engagement Methodology

2.1 Sessions Held

Date	Department	IBKY / NKG participants	Format
Mon 5 Oct 2026	Finance	Carol Muthama, Caroline Nzisa	Workshop with live walk-through of the reconciliation files
Tue 6 Oct 2026	Warehouse & Operations	Margaret Mburu, Luis Brian	Workshop with XBS download and app walk-through
Wed 7 Oct 2026	Trade & QC	Jad Belmezouar, Njambi Gatheru, Daniel Magu, Judy Murimi, Billy Rotich	Workshop with live allocation and Comtras walk-through
Wed 7 Oct 2026	Logistics	Whitney Mwangi, Francis Magomba	Workshop

Pathways participants: Daphne Okwaro, Moses Masibo, Denis Otieno, Nicole Akinyi, William Imo, Andrew Tanui, Jones Alukwe.

2.2 Approach

- Live walk-throughs of the Power Apps stock take application, the XBS download/upload process, Comtras and the reconciliation workbooks.
- Process mapping of the as-is stock lifecycle and month-end reconciliation, from period end to sign-off.
- Pain-point identification with business impact, and proposed solutions agreed in the room.
- Requirements elicitation across inventory management, reconciliation, exception handling, reporting, integration and data quality.
- Risk and dependency assessment for the project risk register, and success criteria agreed with stakeholders.
- Review of all four session recordings after the workshops to capture details missed in the notes.

2.3 Reports and Documents

Reports, working files and reference documents used or produced by each department during the stock take and reconciliation. They are needed to complete the requirements and to design the reconciliation solution.

Department	Report / document	Purpose	Status
Finance	Stock take monthly report — XBS	System position at month-end: what XBS says we have	Shown
Finance	Stock take monthly report — Power App	Physical count result per outturn and location: what we counted	Shown

Department	Report / document	Purpose	Status
Finance	Comtras/BRIC inventory report (526) and sales relief (537)	Basket-level position, contract balances, inventory incl. stuffed-not-invoiced, and sales relief — the third leg of the reconciliation	Shown — extract to be shared
Finance	Three-way reconciliation report	Physical vs XBS vs Comtras positions and variances	Shown
Finance	Reconciliation workbooks with variance explanations (P04 and recent months)	Level 1 and Level 2 reconciliation and reasons; context for the AI assistant	To be shared
Finance	Shorts-in-transit report	Contracted vs delivered weight differences	Requested
Warehouse & Ops	XBS stock download (current, processing, WIP, dispatch)	Source for the app upload and stock categories	Shown
Warehouse & Ops	Ex-office / sweepings register (weekly)	Stock held outside XBS	Requested
Warehouse & Ops	Grade and quality classification reference	Grade to quality code reference	Outstanding
Trade & QC	XBS — All Party Allocation report	How purchased bags are split across destinations and allocated to a sales contract: sales contract, feeding purchase contracts, bags on contract, bags to allocate, balance	Shown
Trade & QC	Allocation — Comtras format	Same allocation reworked for Comtras/BRIC; the bridge between lot-level and basket-level allocation	Shown
Trade & QC	Quality long/short report (manual, maintained by QC)	Hold vs sold by quality code: what we have (long) vs what we owe (short)	Shown — to be shared
Trade & QC	Comtras long/short report	Commercial long/short at basket and contract level	Shown
Trade & QC	Direct purchase overview	Master tracker of direct purchases from contract to payment; source of direct-purchase variance	Shown — to be shared
Trade & QC	Secondary-to-primary basket mapping (incl. Low Grade)	Basis of the master mapping	Outstanding
Trade & QC	Auction and direct catalogues	Duplicate-offer checks	Requested
Logistics	Third-party stock statements and pre-count confirmation letters	Third-party balances before the count	Requested
Logistics	Third-party ageing report (0–90 days)	Coffee held too long at third parties	Requested
Logistics	Shipped-during-month report	Explains movements around the count date	Requested

2.4 Session Outputs

- As-is process narratives with step-by-step flows per department (Sections 4–8).
- Pain-point registers with business impact and proposed solutions.
- Consolidated, numbered requirements, exception catalogue and report list (Sections 9–12).
- Open questions, decisions and action items with owners (Sections 15–16).

3. Key Concepts and Definitions

These definitions were confirmed or used in the sessions. They are the shared vocabulary for design and testing.

Term	Definition
Outturn	Batch identifier given to a lot of coffee at milling (mill initials, sequence, week). The main key in XBS and the app.
Lot / sale lot	An outturn as offered at auction (sale number and lot number). One outturn can be split across lots and locations.
Warrant	Ownership document issued by the licensed warehouseman once coffee is paid for. Its quantity can differ from the contract and from what is delivered.
Grade / quality code	Physical grade (e.g. AA, AB) and the quality code from green and cup analysis.
Secondary basket	Detailed quality bucket assigned to each lot. Many secondary baskets roll up to one primary basket.
Primary basket	Commercial bucket used in Comtras/BRIC and contracts: Plus, FAQ, Grinder and Low Grade (recently created for processed coffee that does not qualify as Grinder).
Bag	A full 60 kg bag (standard packing).
Pocket	A part-filled bag, recorded by weight (anything under 60 kg).
Unit	Number of physical sacks: full bags plus pockets (e.g. 53 bags + 1 pocket = 54 units).
60 kg equivalent	Total weight ÷ 60. Used to compare positions across systems.
Lights / rejects	Light processed coffee: about 50 kg fills a 60 kg bag. Dividing weight by 60 understates bags (1,000 kg = 16 bags + a 40 kg pocket = 17 units).
Current stock	Coffee in XBS available at Ibero or third-party warehouses.
Processing stock	Lots blocked in XBS for a processing instruction.
Work in progress (WIP)	Coffee physically processed or dispatched where one system step is still missing in XBS.
Dispatch / stuffed	Coffee loaded into containers. It leaves XBS at stuffing but stays in BRIC inventory until invoiced.
Stuffed-not-invoiced	Coffee stuffed out (on the way to ICD, port or at sea) but not yet invoiced; a standard reconciling item between XBS and Comtras.
Provisional stock	Direct-purchase coffee contracted but not yet received; tracked from contract → warrant → delivery.
Direct-purchase stock	Coffee bought directly from growers/cooperatives under a direct contract rather than at auction.
Franchise	Volume booked as stuffed but physically left behind (about 1% of a stuffing), accumulating over time.
Ex-office	Coffee from quality samples, grouped by quality category and later bulked back into stock.
Sweepings	Coffee recovered from sweeping and cleaning, grouped and later reintegrated.

Term	Definition
Loose bags	Partially filled or loose quantities not attached to a full lot.
Disputed stock	Stock under query (e.g. short delivery owed by a third party).
Long / short	Long = coffee owned (in stock or bought); short = coffee sold forward but not yet covered.
Short in transit	Weight difference between what was contracted/dispatched and what arrived (e.g. 100,000 kg contracted, 98,000 kg received).
526 report	Comtras/BRIC inventory report used by Finance (includes stuffed-not-invoiced coffee).
537 report	BRIC sales report that relieves inventory against primary basket and sales contract.
Lot blocking	Allocating specific lots to an export contract in XBS.
Cut-off	The point after which no further postings are included in the month-end position.
Data correction	Administrative correction in XBS. XBS is not a financial ledger; financial adjustments are posted in BRIC/Sun.

4. End-to-End Stock and Reconciliation Cycle

4.1 Stock Lifecycle

Stage	What happens	Owner	System
Offer	Broker catalogue (auction) or direct offer received; samples analysed	Trade, QC	Excel, XBS catalogue
Purchase	Auction on Tuesday, confirmed Wednesday; direct contract per deal	Trade	NCE, Comtras, XBS
Payment and warrant	Auction paid within about a week; direct paid on delivery; warrant issued	Finance, Trade	Bank, Sun, XBS
Contract allocation	Bags added to purchase contracts per primary basket in Comtras/BRIC (100-amendment limit)	Trade	Comtras
Movement	Movement instruction; collection from third-party warehouse; may be partial	Logistics	XBS, email
Receipt	Weighbridge, receiving ticket, QC acceptance	Warehouse, QC	Weighbridge, XBS
Storage	Ibero or third-party; locations; multi-location lots	Warehouse	XBS (partly manual)
Processing	Sorting, regrading, bulking; outputs move to new secondary baskets or Low Grade	QC, Warehouse	XBS
Allocation and stuffing	Lots blocked to sales contracts; bagging/stuffing instruction; container loaded	Trade, QC, Warehouse	XBS
Shipment and invoice	ICD, sailing, documents; BRIC invoiced; 537 relieves inventory	Logistics, Trade, Finance	XBS, BRIC

4.2 Month-End Calendar

When	Activity	Owner
Last 3 days of month	Posting cut-off: all purchases, receipts, quality releases, processing results and transfers posted in XBS	All departments
Day 1	XBS stock downloaded as at the last day of the month and uploaded to the app; this extract is the reference for the count	Warehouse & Operations
Days 1–3	Physical count at Ibero and third-party warehouses (about 3 standard days)	Operations
Count + 1	Level 1: physical vs XBS by outturn and location; recounts; explanations	Operations, Finance (Caroline)
Count + 2 to 5	Level 2: XBS vs Comtras/BRIC by basket and contract; reconciling items	Finance, Trade
By working day 7	Post-stock-take review and electronic sign-off (SOP target)	All signatories

Counts take place every month; a 100% count is done twice a year (December and June) during the auction recess.

4.3 Two Reconciliation Levels

Level	Compares	Typical reconciling items
Level 1	Physical count (app) vs XBS, by outturn, location and grade	Direct sales after the count; data corrections not yet posted; coffee found in another warehouse; multi-location lots; unit/pocket differences; lights; third-party stock delivered after the count
Level 2	XBS vs Comtras/BRIC, by secondary and primary basket and contract	Stuffed-not-invoiced; processing and downgrades not reflected in Comtras; rolled-over contracts; direct purchases (provisional) not yet in XBS; franchise, ex-office, sweepings, loose bags; credit notes not posted; auction purchases not yet uploaded

4.4 Unit-of-Measure Rules

- Positions are compared on weight and on 60 kg equivalents (weight ÷ 60); units are reported separately.
- A pocket counts as one unit; its weight is recorded separately. The app and XBS must apply the same rule (to be confirmed in XBS).
- Lights are packed in 60 kg bags but weigh about 50 kg. Packing type must drive the conversion so that lights are not understated when divided by 60.
- Allocation in XBS is ratio-based and can leave fractional remnants (e.g. 0.001 bags, 1–8 kg) on old lots; these need a rounding and clean-up rule.

5. Finance

5.1 Role

Finance owns the certified month-end stock position. It runs the Level 2 reconciliation against BRIC/Comtras, reviews the Level 1 result, chairs the post-stock-take review and signs off. Finance is accountable for the reconciled numbers even though it does not own the data in XBS.

5.2 As-Is Process

Step 1 — Cut-off and extract

All teams post outstanding transactions in the last three days of the month. On the first day of the new month the warehouse team downloads XBS as at the month-end date and uploads it to the app. The upload is manual: someone downloads the file to a drive and uploads it into the application.

Step 2 — Physical count

Warehouse & Operations complete the count in the Power Apps stock-take application. Counters select the outturn and enter the quantity; results are captured per counter, time-stamped and tied to counter identity, and the app warns when an entry does not match XBS. The app replaced paper count sheets, which removed transcription errors and the roughly five-day wait for paper results to be typed into Excel. The app produces a count result but does not reconcile.

Step 3 — Level 1 reconciliation (physical vs XBS)

Operations perform an initial physical-vs-XBS reconciliation in Excel and produce a reconciliation pack (for April 2026, a 15,500-row, 2.3 MB workbook). Finance (Caroline) compares the count with XBS at outturn and grade level, in 60 kg equivalents, using a unique reference per line. Outturns found physically but not in XBS, and in XBS but not found, are investigated. Once the data is clean this check takes under an hour.

Example discussed (P04, approximate figures — to confirm): a variance of about 2,000 bags between XBS and the physical count was explained by about 770 bags of direct sales after the count, data corrections such as a credit note not yet posted, coffee awaiting re-assignment to another outturn, and about 1,000 bags at KPCU delivered after the count.

Step 4 — Level 2 reconciliation (XBS vs BRIC/Comtras)

Finance rebuilds the position against BRIC stocks and longs using the Comtras/BRIC 526 inventory report. BRIC treats stuffed-out coffee as inventory until it is invoiced, so the stuffing-out report from XBS is added back. Direct purchases committed in BRIC but not yet in XBS, processing and downgrades, and rolled-over contracts must then be traced by hand. This takes about two weeks and is the main cause of the 33-day cycle.

Step 5 — Post-stock-take review

A cross-functional meeting reviews the variance report. It typically triggers fresh Excel forensics on direct sales, broker stocks and AGL, and a further variance-review pack is prepared and circulated. This can repeat several times as new variance theories emerge.

Step 6 — Final report and sign-off

Once Finance is satisfied with the explanations, the final reconciliation report is circulated (for April 2026, on 2 June — 33 calendar days after period end). The reconciliation is printed and signed on paper by Finance, Operations, Trade, Warehouse and Logistics, with no approval timestamps, no visibility of outstanding approvals and no drill-down to supporting data.

Adjustments and data corrections

XBS is a warehouse system, not a financial ledger. Corrections in XBS are administrative (data corrections). Financial adjustments such as credit notes are posted in BRIC/Sun. No adjustment is made without review and approval.

Payment timing

- Auction coffee is paid within about a week of the auction; warehouse charges start after that week. Licensed warehousemen guarantee that auctioned coffee exists.
- Direct-purchase coffee is normally paid on delivery. Exceptions where payment preceded delivery have occurred.

5.3 Reconciling Items Observed

Item	Description	Typical owner
Sales after the count	Direct sales or shipments after the count date but within the period	Trade
Data corrections pending	Corrections or credit notes agreed but not yet posted in XBS/BRIC	Trade, Finance
Late third-party deliveries	Coffee delivered after the count (e.g. KPCU)	Logistics
Stuffed-not-invoiced	Coffee out of XBS but still in BRIC inventory	Trade, Logistics
Direct purchases in transit	Contracted/warranted but not delivered	Trade
Shorts in transit	Weight lost between dispatch and receipt (e.g. 100,000 kg vs 98,000 kg)	Logistics
Processing and downgrades	Basket changes not reflected in Comtras	QC, Trade
Non-XBS categories	Franchise, ex-office, sweepings, loose bags	Warehouse, QC

5.4 Pain Points and Proposed Solutions

ID	Pain point	Business impact	Proposed solution (requirement ref.)
F1	Manual XBS download and upload	Time-consuming, error-prone; different files for the same date	Automated ingestion of XBS via Power Automate and dataflows into the governed data warehouse. Until APIs are available, use controlled Excel/CSV extracts on a schedule; design for future API integration (FR-01, FR-02).

ID	Pain point	Business impact	Proposed solution (requirement ref.)
F2	No single source of truth; reports for the same date return different balances	Users work from different balances; reconciliation disputes and rework	One certified data model in the Azure Data Warehouse, with period-end snapshots and period locking so historical balances do not change after later processing (FR-03, FR-04).
F3	Level 2 traced by hand	About two weeks per cycle; key-person dependency	Master mapping and automated XBS vs Comtras/BRIC reconciliation with reconciling items (FR-10, FR-41, FR-42).
F4	Variance shows what, not why	15–20 person-days per cycle of forensic Excel; explanations rebuilt every month	Exception register with root-cause categories (duplicate entry, unposted delivery, mapping error, part lot, cancelled contract, direct sale after count, master-data clean-up, etc.) and evidence; Copilot Studio assistant for interpretation and next-best action, advisory only (FR-50–FR-53, FR-80).
F5	Direct purchases tracked in Excel	Contracted vs warranted vs delivered not visible; risk of double purchase	Provisional stock tracking with quantity chain and duplicate checks (FR-20, FR-60, FR-61).
F6	Lights and pockets distort comparisons	False variances in 60 kg equivalents	Packing-type-driven conversion rules shared by app and reports (FR-11).
F7	No real-time variance tracking or daily view of exceptions	Issues found only at month-end	Real-time Physical vs System vs Variance dashboard with drill-down by warehouse, location, lot, grid and grade; exceptions raised on the day and pushed to owners via Teams/Outlook (FR-30, FR-52, FR-54).
F8	Explanations not reusable	AI assistant has no context	Store historical explanations by outturn and category as context for the Copilot assistant (FR-80, FR-81).
F9	Paper sign-off	No audit trail, no timestamps, no visibility of outstanding approvals	Digital reconciliation workflow with electronic sign-off, owner assignment, review status and SOP-timeline tracking; sign-off pack in Excel/PDF to a distribution list (FR-70, FR-71).
F10	Device and network issues at third-party sites	Lost count time	Device readiness check and offline-tolerant capture where needed (NFR-05).
F11	Limited visibility of WIP and other stock categories	Incomplete inventory; reconciling items without a system of record	Capture WIP, processing, provisional, direct-purchase, sweepings, ex-office, disputed, loose bags and franchise as separate categories, included in reconciliation and variance reporting (FR-20–FR-23).
F12	Third-party warehouse reconciliation challenges	Extra reconciliation effort	Third-party control process: standard report format, request balances, compare to XBS and resolve exceptions before the count (FR-65).
F13	Stock movements hard to trace	Slow investigations and audits	Location management by warehouse, zone, grid, location and lot; every movement linked to a unique reference (FR-34).

6. Warehouse & Operations

6.1 Role

The warehouse is the physical custodian of green coffee from the gate to the container. It sits between Logistics (which delivers coffee from third-party or auction warehouses) and Quality Control (which inspects and formally accepts coffee into the system). Operations runs the stock take: it downloads the XBS position, sets up the count, assigns counters, runs count day and performs the Level 1 clean-up.

6.2 As-Is Process

Stock categories in the XBS download

Category	Meaning (as explained)
Current stock	Coffee available at Ibero and third-party warehouses (AGL, Ibero Export, KCC, Maxa, Mitchell Cotts, KPCU and others)
Processing	Lots blocked for a processing instruction
Work in progress	Coffee processed or dispatched where one step is still missing in XBS
Dispatch	Coffee stuffed out

XBS fields used: item name, item group, item form, secondary basket, item place, inventory type, crop season, store/warehouse. Operations would like the app to fetch these categories directly instead of downloading files.

Receiving, storage, processing and release

The department runs two systems — XBS and a separate weighbridge system — that are not integrated. This disconnect is the main source of manual duplication in the daily workflow.

Phase A — Pre-arrival

- A third-party service provider supervises loading of trucks at third-party storage. The transporter or third party sends a WhatsApp message to the warehouse with what was loaded, which lots and how many bags.
- The warehouse confirms readiness and finds floor space informally; there is no system pre-arrival notification or checklist. Logistics issues transport instructions by phone or message.

Phase B — Gate and weighbridge

- Security checks driver identity, documents, truck details, delivery notes and PPE, and records visitor access.
- Gross weight is recorded on the weighbridge, then read and retyped into XBS to open the receiving ticket.

Phase C — Receiving and ticket management

- The receiving ticket in XBS is the control point. Bags are recorded load by load against the open ticket.
- Tare weight is captured; net weight (gross minus tare) is reconciled against the recorded loads. A mismatch triggers a manual, line-by-line investigation of the stock transfer receipt.

- The ticket is closed when the weights agree and quantity ownership passes to the Warehouse. QC then maps the ticket and formally accepts the quality; until then the coffee is not available for processing or shipment.

Phase D — Storage and location

- Put-away locations are chosen by walking the floor; there is no system location view. Positions are tracked informally and internal moves are not always recorded in XBS in real time.
- The same outturn can be stored in several locations or warehouses (e.g. different AGL sites).
- Part of the stock stays at third-party storage; visibility depends on manual communication.

Phase E — Processing

- QC creates the processing instruction in XBS (warehouse, type, lots). The warehouse prints it as a job card.
- Processing (colour sorting, gravity separation, regrading, bulking) produces different outputs: clean coffee vs rejects, bolt vs lights, above-screen vs below-screen.
- Results are posted to XBS afterwards from the job card; QC is not notified automatically and follows up by phone. QC reviews basket transitions; a second instruction is treated as a process failure needing root-cause investigation.

Phase F — Release and shipment

- QC allocates lots to export contracts (lot blocking). The warehouse loads containers; seals are verified but the system does not show seal issuance or changes.
- The loaded container is weighed at the exit weighbridge, with the same manual retyping as at intake.

Stock take — before the count

- XBS extract downloaded and uploaded to the app at the agreed date; the shared file becomes the reference for all teams.
- Third-party balances requested through Logistics and compared with XBS.
- Counter briefing; counters assigned to locations.

Stock take — on count day

- Counters select the outturn and enter bags, pocket weight and packing type; the app flags mismatches with XBS.
- The current dashboard only shows items that exist in XBS. Operations needs to see, live: counted vs not yet counted, counted but not in XBS, and in XBS but not counted.
- When coffee is found in a different warehouse from XBS (e.g. at Maxa while XBS says AGL), the counter must not be blocked; the app should record where it was found and propose a data correction.
- When the same lot is picked up by different counters or in different locations, the app should warn and show where it was already counted.

Stock take — after the count

- Variances reviewed and recounts done; reasons recorded.
- Ex-office and sweepings are counted against their own register (updated weekly), not against XBS. When they are put back into XBS as current stock, the quantity is released from the register so the balance stays correct.

6.3 Reconciling Items Observed

Item	Description	Typical owner
Coffee found in another warehouse	Counted at a different warehouse or location from XBS (e.g. Maxa vs AGL)	Warehouse, Logistics
Duplicate and multi-location lots	Same outturn stored or counted in several locations or warehouses	Warehouse
Internal moves not recorded	Bags moved within the warehouse without an XBS update	Warehouse
Processing not yet posted (WIP)	Processing done on the floor but results not yet posted from the job card	Warehouse, QC
Received but not QC-accepted	Receiving ticket closed but QC ticket mapping and acceptance pending	Warehouse, QC
Receiving weight variance	Net weighbridge weight differs from the loads recorded on the ticket	Warehouse
Unit, pocket and lights differences	Pockets, lights and loose bags converted differently in the app and XBS	Warehouse, Finance
Stock held outside XBS	Ex-office, sweepings, loose bags and franchise counted against their own registers	Warehouse, QC
Third-party stock not yet moved	Coffee still at third-party storage or in transit at count date	Logistics, Warehouse

6.4 Pain Points and Proposed Solutions

ID	Pain point	Business impact	Proposed solution (requirement ref.)
W1	Different XBS downloads for the same date	Conflicting stock positions	Automated, single controlled extract at cut-off with standard timing and content; period-end snapshot (FR-01–FR-03).
W2	Dashboard shows only XBS items	Found-not-in-XBS and not-counted items invisible during the count	Live count dashboard with three views and drill-down (FR-30).
W3	Duplicate or multi-location counting	Inflated counts; failed reconciliation	Detect duplicate lot numbers, duplicate counting and lots spread across locations; prompt the counter with where it was already captured; record with a flag rather than block (FR-31).
W4	Coffee found in a different warehouse	Counters blocked or forced into manual entry	Capture the actual location and raise a data-correction exception (FR-32).

ID	Pain point	Business impact	Proposed solution (requirement ref.)
W5	Ex-office, sweepings and franchise not tracked	Incomplete stock; missed reconciliation items	Category registers with add, count, remove and release; compare sweepings with operational records; remove ex-office from current stock when transferred; keep quality per category; structured reintegration into XBS (FR-21, FR-22).
W6	Lights and pockets handled inconsistently	Over- or under-statement in 60 kg equivalents	Packing-type rules (FR-11).
W7	Wrong warehouse at XBS upload	Reconciliation failures	Validate warehouse assignments at ingestion and upload against the master mapping, warrant and movement instruction; flag mismatches (FR-62).
W8	No processing-complete notification	QC follows up by phone and checks XBS manually	Power Automate notification to QC (Teams/email) when processing results are posted (FR-55).
W9	Counter capability varies	Errors and slow counts	Counter training and quick-reference guide before each count (FR-90).
W10	Weighbridge not linked to XBS	Double entry at receipt and dispatch; error risk; slower tickets	Weighbridge-to-XBS connector following the Tanzania model (Uganda is building the same). Outside the current approved scope; logged for the roadmap (Section 13).
W11	Loose and partial bags not systemised	Under- or over-stated stock	Capture loose and partial bags separately, convert to equivalent units, keep an audit trail and include them in variance calculations (FR-21, FR-11).
W12	No historical snapshots	Audit challenges	Retain period-end snapshots in the data warehouse for audit retrieval and trend analysis without post-period change (FR-03, NFR-08).

7. Trade & Quality Control

7.1 Role

Trade buys coffee through two channels — auction via Ibero Kenya and direct purchases through BRIC — and owns contracts, pricing, hedging and shipment allocations. Quality Control manages sample intake, green analysis and cupping, basket classification, processing instructions, lot allocation to export contracts, and pre-shipment and shipment samples. Both work heavily in XBS (purchases, warrants, processing, contract mapping) and in Comtras/BRIC (basket positions, contract balances, sales relief).

7.2 As-Is Process

Auction purchases

Brokers send catalogues before the auction with lot numbers, grades, special remarks, number of units, bags and pockets. QC performs green analysis and cupping on all lots and produces a quality report for traders. Traders bid on Tuesday and confirm on Wednesday. By Thursday or Friday Trade instructs BRIC to add the bags to purchase contracts, per primary basket (for example, one active contract for FAQ).

Direct purchases and contract timing

A new contract is created for each direct purchase. The quantity changes at each stage:

Stage	Example (bags)	Effect
Contracted	1,000	Committed volume booked in Comtras/BRIC as a long
Warranted	900	XBS receives 900; Comtras still shows 1,000 until amended
Delivered	800 (partial)	Received as current stock; Finance/Trade follow up the 100-bag balance
Confirmed undeliverable	—	Contract amended to the delivered quantity; payment on delivered quantity

Comtras balances can therefore temporarily exceed XBS, and XBS can exceed what is physically delivered and counted. Short deliveries are common, especially on direct purchases; some balances are owed by the third-party warehouse. Comtras treats direct coffee as a long while it is at a third-party warehouse and as inventory once invoiced into BRIC.

Contracts and roll-over

- A contract in Comtras accepts at most 100 amendments. When the limit is reached, BRIC emails Trade, Trade creates a new contract and the balance is transferred. The old contract keeps a residual balance.
- When Finance downloads the 526 report, contract balances do not match XBS without manually tracing rolled-over contracts and their originating balances.
- XBS does not accept duplicate contract numbers, so Trade adds the sale number and season to make allocations unique.
- Roll-over coordination is by email; balances must be traced across contract generations.

Baskets, processing and downgrades

- Processing does not upgrade coffee; it separates lower-quality material already in the lot. Outputs (clean coffee and rejects/lights) keep the same primary basket but get new secondary baskets, so post-processing stock no longer maps one-to-one to the original outturn — it has been split, bulked or regraded.
- A fourth primary basket, Low Grade, was created for processed coffee that does not qualify as Grinder. Primary baskets are now Plus, FAQ, Grinder and Low Grade.
- Processing loses about 3% of volume. If downgrades are booked only when low-grade coffee is shipped at season end, a large loss lands at once (about USD 600,000 was mentioned — to confirm). Trade plans to move processed downgrades to Low Grade monthly to spread the loss.
- Until this is posted consistently, the Plus basket in Comtras holds more coffee than physically exists, because amendments have not followed processing.

Dispatch, allocation, stuffing and invoicing

- Trade identifies the lots to ship and performs lot blocking in XBS. A bagging instruction is the last instruction before coffee leaves the warehouse.
- XBS sends a notification when stuffing happens; Trade downloads the allocation report.
- Trade fixes ratio-based fractions (e.g. 0.001 bags; 1–8 kg remnants from lots more than a year old), filters by seller (Ibero) and primary basket, and maps each line to the active purchase contract.
- A pivot by sales contract and purchase contract is built and reformatted for BRIC and sent by email.
- BRIC invoices; the 537 report relieves inventory. A shipment can combine auction (Ibero) and direct coffee; the seller field in XBS shows the split.
- Coffee leaves XBS at stuffing but stays in BRIC inventory until invoiced. Stuffing to ICD is about 10 days; documents follow about 5 days after sailing. Long delays tie up working capital. The eDocs initiative will generate shipping documents directly from the shipping instruction.

Three positions

Position	Source
XBS long	Actual stock by outturn and secondary basket (purchases, warrants, processing, stock)
Comtras long/short (526)	Basket and contract balances incl. stuffed-not-invoiced and shorts
Quality long/short (manual)	Excel maintained by QC (Billy Rotich): XBS position by quality vs forward sales; shorts are built by hand because sales are allocated to one secondary basket but shipped as blends

Example shown (dates not aligned — to confirm): FAQ about 40,000 bags in one view vs 22,000 in the other; Grinder about 16,000 vs 7,000; Plus about 1,400 vs 6,500; totals about 31,000 vs 64,000. The positions must be compared at the same cut-off and the differences explained by the reconciling items in Section 5.3.

Coffee outside XBS

Category	How it arises	Requirement
Franchise	About 1% of a stuffing booked as stuffed but left behind; accumulates (about 500 bags)	Franchise register; quality taken from the stuffing contract
Ex-office	Samples kept after analysis, grouped into quality categories 1, 2 and 3, later bulked into XBS and Comtras	Register with quality category and original source
Sweepings	Recovered from cleaning; grouped	Register; reintegration workflow
Loose bags	Partial or loose quantities	Register; conversion to equivalent units

Today these are reintroduced as symbolic direct purchases at a nominal price. The requirement is to keep a register, track increases and decreases, keep the original quality, and release quantities from the register when reintroduced into XBS.

Provisional stock (agreed proposal)

Trade, QC and Finance supported a provisional-stock model: committed direct purchases are held as provisional stock at contracting, confirmed against the warranted volume when warrants arrive, and moved to current stock on physical receipt. Provisional stock is included in counts so the physical count and the system position align before delivery is complete. Trade also suggested recording offers from the direct catalogue as "offered" before purchase. Where provisional stock is held (an XBS module or the new solution) is a decision for IT and Trade.

Duplicate offers and warrant location

- The same outturn can be offered directly and then at auction. Trade wants an automatic check of direct and auction catalogues on upload, with an alert listing duplicates.
- Lots not confirmed after auction are sometimes re-offered later; tracking re-offers is useful but not a priority.
- Warrants sometimes name AGL while the coffee is at Maxa or Mitchell Cotts. Change instructions are issued, and a second instruction is needed to move the coffee to Ibero.

7.3 Reconciling Items Observed

Item	Description	Typical owner
Stuffed-not-invoiced	Coffee out of XBS at stuffing but still in BRIC inventory until invoiced	Trade, Logistics
Direct purchases not yet delivered	Contracted or warranted volume not yet received (provisional stock)	Trade
Short deliveries	Delivered below warranted or contracted quantity; balance owed	Trade, Logistics
Processing and downgrades not posted	Basket changes (incl. Low Grade) not reflected in Comtras	QC, Trade

Item	Description	Typical owner
Rolled-over contract residuals	Balances left on old contract numbers after the 100-amendment roll-over	Trade, Finance
Auction purchases not yet uploaded	Bought at auction but not yet added to contracts or uploaded to XBS	Trade
Data corrections and credit notes pending	Agreed corrections not yet posted in XBS or BRIC	Trade, Finance
Fractional remnants	Ratio-based allocation remnants (e.g. 0.001 bags, 1–8 kg) on old lots	Trade
Franchise and ex-office	Stuffed-but-left-behind volume and retained samples outside XBS	Trade, QC

7.4 Pain Points and Proposed Solutions

ID	Pain point	Business impact	Proposed solution (requirement ref.)
T1	No master mapping across systems	Level 2 rebuilt by hand	Master mapping incl. Low Grade (FR-10).
T2	Allocation report reformatted by hand	Rework every shipment	Allocation report generated once in BRIC/internal format from XBS data, rounding rules applied, distributed automatically (FR-63).
T3	Contract roll-over by email; 100-amendment limit forces new contract numbers	Residual balances hard to trace; reconciliation gaps	Contract lineage (old → new) in the data warehouse, shown in the reconciliation dashboard; automated alert near 100 amendments (FR-64).
T4	Downgrades not posted consistently	Comtras overstates higher baskets	Monthly downgrade worksheet and basket-transition tracking (FR-12).
T5	Three positions not aligned; Comtras balances do not match XBS	Disagreement on the stock figure	XBS vs Comtras vs quality long/short side by side at one cut-off, with the difference broken down by reconciling item (FR-42).
T6	Stuffed-not-invoiced not aged or explained	Manual ageing each cycle; working-capital impact	Standard reconciling item with ageing, contract reference, expected invoice date and threshold alerts; link to shipping-document generation to shorten stuffing-to-invoice (FR-43).
T7	Purchase, warrant and delivery quantities differ	Contract balances do not match stock or XBS	Provisional stock and quantity chain; provisional stock included in counts (FR-20, FR-60).
T8	Duplicate offers	Risk of paying twice for the same coffee	Catalogue duplicate check with alert (FR-61).
T9	Franchise, ex-office, loose bags and sweepings not in the application	Reconciling items without a record	Category registers with increases, decreases, quality and a release function on reintegration (FR-21, FR-22).

ID	Pain point	Business impact	Proposed solution (requirement ref.)
T10	Fractional remnants on old lots	Noise in allocation and reconciliation	Rounding and clean-up rule; remnant report (FR-13).
T11	Processing creates new secondary baskets under the same primary basket	Post-processing stock does not map one-to-one to original outturns	Mapping of secondary-basket transitions per processing instruction, surfaced in the QC review (FR-12, FR-14).
T12	No real-time XBS vs Comtras reconciliation	Variance found at month-end, not on the day	Daily XBS vs Comtras comparison in Power BI; exceptions flagged on the day (FR-42, FR-52).
T13	Quality results and basket classification not visible in the stock application	QC reconciles basket transitions manually	Show basket classification, secondary-basket transitions and quality results in the application for QC review (FR-14).

8. Logistics

8.1 Role

Logistics makes sure coffee is where the records say it is. It moves coffee into and out of Ibero warehouses, makes sure coffee is received at the correct warehouse (Ibero or third-party), coordinates movements, receipts and dispatch with third-party operators, and gives status updates on coffee in transit and in storage. Logistics does not own the physical stock or the system of record. During the stock take it confirms which lots have moved and which remain at third parties, and it is usually asked to explain differences between the system and what was found.

Logistics has no system view of third-party positions; communication relies on WhatsApp, email and phone calls.

8.2 As-Is Process

Before the count

- Logistics writes to each third-party warehouse with the count schedule and the stock IBKY believes is held there.
- Third parties are asked to confirm or send their balances. Only two of five do so consistently; reminders by email and phone often go unanswered.
- Logistics aims to have coffee collected before the count; partial collections spill over (e.g. 50 of 100 bags collected, the rest remains until after the count).

After the count

Logistics is asked to explain differences between XBS and what was found, mainly by phone and email. The typical variances are listed in Section 8.3.

Needs raised in the session

- A comment section on each variance, with owner and resolved status, so explanations are recorded once — replacing follow-up calls and emails — and can be reused by the AI assistant.
- An ageing view of third-party stock (0–90 days), with reminders before limits are reached (15 and 20 days were suggested; SOP target 21 days).
- Data corrections batched into one clean-up rather than one at a time.
- A report of what was shipped during the month.

8.3 Reconciling Items Observed

Item	Description	Typical owner
Partial collection	Part of a lot at Ibero, the rest still at the third-party warehouse	Logistics
Wrong warehouse at upload	Coffee uploaded as AGL but held elsewhere	Logistics, Warehouse
Uploaded only in XBS	Outturn in XBS but not yet moved or received	Logistics
Weight loss / short delivery	Bought 100 kg, received 98 kg; common on direct purchases	Logistics, Trade

Item	Description	Typical owner
Late movement	Coffee moved after the count date	Logistics
Grade mismatch	Grade recorded differs from the grade found	Logistics, QC

8.4 Pain Points and Proposed Solutions

ID	Pain point	Business impact	Proposed solution (requirement ref.)
L1	Third parties do not return balances	Unconfirmed stock at count	Standard confirmation template and tracker; escalation when not returned (FR-65).
L2	Logistics repeatedly asked to explain misalignments (e.g. grade mismatches) by phone and email	Disruption to other work; explanations lost in email threads	Comment and resolution log on each exception, recorded directly in the app (FR-53).
L3	Third-party ageing not monitored	Storage costs; stock at risk	Ageing dashboard and alerts (FR-66).
L4	Partial collections across the count date	Interim variances	Part-lot list before the count; in-transit status (FR-33).
L5	Many single data corrections	Admin effort	Batched correction list for one clean-up (FR-56).

9. Consolidated Business Requirements

Priority: **Must** = needed for go-live; **Should** = high value, planned for the MVP if capacity allows; **Could** = later phase. Phase refers to the delivery roadmap in the approved business case (1 = reconciliation foundation, 2 = visibility, 3 = AI-assisted). Priorities and phases are Pathways proposals to be agreed with the business owner.

9.1 Data, Cut-off and Snapshots

ID	Requirement	Priority	Source	Phase
FR-01	Load the XBS stock position (current, processing, WIP, dispatch, with item, basket, inventory type, crop season and warehouse fields) from a controlled extract at the agreed cut-off; no manual re-keying.	Must	Finance, Ops	1
FR-02	Load Comtras/BRIC 526 inventory, 537 sales and contract balances at the same cut-off.	Must	Finance, Trade	1
FR-03	Store a locked month-end snapshot of all positions; later postings must not change the certified position. Retain snapshots for audit and trend analysis.	Must	Finance	1
FR-04	Record the extract time and file used for every count, so all teams work from the same data.	Must	Ops	1
FR-05	Review and approve each loaded extract (row and weight control totals) before it becomes the period snapshot (data ingestion approval).	Should	Finance, Ops	1

9.2 Master Data and Conversion Rules

ID	Requirement	Priority	Source	Phase
FR-10	Maintain a master mapping: outturn → grade/quality code → secondary basket → primary basket (Plus, FAQ, Grinder, Low Grade) → contract, with effective dates and Finance approval of changes.	Must	Finance, Trade, QC	1
FR-11	Apply one set of conversion rules (packing type incl. lights at about 50 kg, pocket = 1 unit, loose bags to equivalent units, 60 kg equivalents) in the app, reconciliation and reports.	Must	Finance, Ops	1
FR-12	Track basket transitions from each processing instruction and the monthly downgrade to Low Grade, so XBS and Comtras can be aligned.	Must	Trade, QC	1
FR-13	Apply a rounding rule to ratio-based allocations and report fractional remnants on old lots for clean-up.	Should	Trade	1
FR-14	Show basket classification, secondary-basket transitions and quality analysis results in the stock application for QC review.	Should	QC	2

9.3 Stock Categories

ID	Requirement	Priority	Source	Phase
FR-20	Track provisional (direct-purchase) stock from contract → warrant → delivery, with quantities at each stage; include it in counts and reconciliation.	Must	Finance, Trade	1
FR-21	Maintain registers for franchise, ex-office, sweepings, loose bags and disputed stock: add, count, adjust and release when reintroduced into XBS, with quality category, original source and audit trail.	Must	Trade, QC, Ops	1
FR-22	Count ex-office and sweepings against their own register and show them as separate categories in the reconciliation.	Should	Ops	1
FR-23	Show WIP, processing and direct-purchase stock as separate categories in the count, reconciliation and variance reporting.	Must	Finance, Ops	1

9.4 Count Execution

ID	Requirement	Priority	Source	Phase
FR-30	Live count dashboard showing counted vs not yet counted, counted but not in XBS, in XBS but not counted, and progress — with drill-down by warehouse, location, lot, grid and grade.	Must	Ops, Finance	2
FR-31	Warn when the same lot is counted twice or in several locations, showing where it was already counted; record with a flag rather than block.	Must	Ops	1
FR-32	Allow counters to record coffee found in a different warehouse from XBS, capture the actual location and raise a data-correction exception.	Must	Ops, Logistics	1
FR-33	Provide a pre-count list of part lots and in-transit coffee by warehouse; track part lots across locations.	Should	Logistics	2
FR-34	Location management by warehouse, zone, grid, location and lot; every stock movement linked to a unique reference for the audit trail.	Should	Ops, Finance	2

9.5 Reconciliation

ID	Requirement	Priority	Source	Phase
FR-40	Level 1: reconcile physical count vs XBS by outturn, location and grade, in weight, units and 60 kg equivalents.	Must	Finance, Ops	1
FR-41	Level 2: reconcile XBS vs Comtras/BRIC by secondary basket, primary basket and contract, adding back stuffed-not-invoiced and provisional stock.	Must	Finance, Trade	1
FR-42	Show XBS, Comtras and the quality long/short side by side at the same cut-off, refreshed daily, with the difference broken down by reconciling item and contract lineage.	Must	Trade, Finance	2

ID	Requirement	Priority	Source	Phase
FR-43	Age stuffed-not-invoiced coffee by contract with an expected invoice date; alert when thresholds are passed.	Should	Trade, Finance	2
FR-44	Report shorts in transit (contracted/dispatched vs received weight).	Should	Finance, Logistics	2

9.6 Exceptions and Explanations

ID	Requirement	Priority	Source	Phase
FR-50	Exception register with ID, type, outturn, warehouse, quantity, owner, department, due date, status and ageing.	Must	All	1
FR-51	Root-cause categories: sale after count, data correction pending, unposted delivery, late delivery, wrong warehouse, duplicate entry, multi-location, part lot, unit/pocket, lights, processing/downgrade, stuffed-not-invoiced, provisional, short delivery, contract roll-over, cancelled contract, mapping error, master-data clean-up — with supporting evidence.	Must	Finance	1
FR-52	Raise exceptions on the day they occur (daily run), not only at month-end; daily exception/complaint log.	Must	Finance	1
FR-53	Comment and resolution log on each exception (who, when, explanation, resolved/unresolved), searchable across months.	Must	Logistics, Finance	1
FR-54	Alerts and escalation through Teams/Outlook when exceptions are raised and when they pass their due date.	Should	All	2
FR-55	Notify QC when processing results are posted.	Could	QC, Ops	2
FR-56	Batch approved data corrections into one clean-up list for XBS.	Should	Logistics, Ops	2

9.7 Purchasing, Contracts and Third Parties

ID	Requirement	Priority	Source	Phase
FR-60	Record the direct-purchase quantity chain and contract amendments; flag contracts where delivered quantity is below contracted.	Must	Trade, Finance	1
FR-61	On catalogue upload, check direct and auction catalogues for the same outturn/grade and send an alert listing duplicates.	Should	Trade	2
FR-62	Validate warehouse and quantity at XBS upload against the master mapping, warrant and movement instruction; flag mismatches.	Should	Ops, Trade	2
FR-63	Generate the allocation report in BRIC/internal format from XBS stuffing data and distribute it automatically.	Should	Trade	2
FR-64	Maintain contract lineage across roll-overs and alert as a contract approaches 100 amendments.	Should	Trade, Finance	2

ID	Requirement	Priority	Source	Phase
FR-65	Third-party confirmation tracker with a standard report format: request, response, balance vs XBS, discrepancies resolved before the count, escalation when no response.	Must	Logistics	1
FR-66	Third-party ageing (0–90 days) with reminders before the 21-day target.	Should	Logistics	2

9.8 Workflow, AI and Enablement

ID	Requirement	Priority	Source	Phase
FR-70	Electronic reconciliation sign-off with approver identity, timestamps, owner assignment, review status, SOP-timeline tracking and audit trail.	Must	Finance	1
FR-71	Sign-off pack in Excel/PDF distributed automatically to an agreed distribution list.	Must	Finance	1
FR-80	Copilot Studio assistant suggests likely root causes and next steps for exceptions, using recorded explanations and historical reconciliations as context; outputs are advisory and subject to human review.	Should	Finance	3
FR-81	Load historical reconciliation explanations (recent months) into a structured form usable by the assistant.	Should	Finance	3
FR-90	SOPs, training material and quick-reference guides for counters, warehouse, Trade, QC and Logistics users; refresher before each count.	Must	Ops	1

10. Exception Catalogue

Owners, timing and thresholds below are Pathways proposals, to be agreed.

ID	Exception	Detection rule	Owner	When
EX01	Found physically, not in XBS	Count line with no XBS record	Ops	Count day
EX02	In XBS, not found	XBS line not counted at close	Ops	Count close
EX03	Wrong warehouse	Counted warehouse ≠ XBS warehouse	Logistics	Count day
EX04	Duplicate / multi-location count	Same outturn counted twice or in several locations	Ops	Count day
EX05	Unmapped outturn	No basket in master mapping	Trade, QC	Daily
EX06	Downgrade not posted	Processing output basket ≠ Comtras basket	QC, Trade	Monthly
EX07	Stuffed-not-invoiced over threshold	Stuffing date + threshold passed without invoice	Trade	Daily
EX08	Short delivery	Delivered < warranted or contracted	Trade, Logistics	On receipt
EX09	Duplicate offer	Same outturn/grade in direct and auction catalogues	Trade	On upload
EX10	Third-party balance not confirmed	No response by deadline	Logistics	Pre-count
EX11	Third-party stock over age limit	Days held > target	Logistics	Daily
EX12	Contract near amendment limit	Amendments ≥ threshold (e.g. 90)	Trade	Daily
EX13	Data correction pending	Agreed correction not posted	Ops, Trade	Daily
EX14	Partial lot count	Lot counted in part at one location with the balance elsewhere or in transit	Ops, Logistics	Count day

11. Reports and Dashboards Required

Report / dashboard	Content	Audience	Frequency
Live count dashboard (Physical vs System vs Variance)	Progress, counted vs not, found-not-in-XBS, in-XBS-not-found, wrong warehouse, quantity mismatches, counting errors; drill-down by warehouse, location, lot, grid, grade	Ops, Finance	Count day (live)
Level 1 reconciliation	Physical vs XBS by outturn, location, grade	Ops, Finance	Each count
Level 2 / three-way reconciliation	Physical vs XBS vs Comtras vs quality long/short with reconciling items and contract lineage	Finance, Trade	Daily view; month-end certified
Exception register	Open, overdue, by owner and root cause, with comments	All	Daily
Stock-category registers	Provisional, franchise, ex-office, sweepings, loose bags, disputed, WIP, processing	Ops, Trade, Finance	Weekly
Stuffed-not-invoiced ageing	By contract, days since stuffing, expected invoice date	Trade, Finance	Weekly
Third-party position and ageing	Balance vs XBS, confirmation status, days held	Logistics	Weekly
Shorts in transit	Contracted/dispatched vs received weight	Finance, Logistics	Monthly
Shipped during the month	Shipments by contract and date	Logistics, Finance	Monthly
Governance KPI scoreboard	Days to sign-off, variance %, exceptions closed on time, same-day posting	Management	Monthly
Sign-off pack	Final reconciliation with approvals	Signatories	Month-end

12. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Security	Role-based access by department and warehouse; separation of duties (preparer cannot approve own reconciliation).
NFR-02	Audit	Every change to counts, mappings, registers, settings and approvals logged with user and timestamp.
NFR-03	Data integrity	Loads reconcile to source control totals (rows and kg); reruns do not duplicate data.
NFR-04	Performance	Count capture supports at least 16 concurrent counters; dashboards refresh within agreed intervals on count day.
NFR-05	Connectivity	Count capture tolerates weak network at third-party sites (to be assessed per site).
NFR-06	Usability	Mobile-friendly screens for counters; plain-language messages; NKG branding.
NFR-07	Ownership	Apps, flows and connections owned by an NKG service account, not personal or external accounts.
NFR-08	Retention	Month-end snapshots retained for audit (period to be confirmed with Finance).
NFR-09	Integration readiness	File-based ingestion designed so that XBS and Comtras APIs can replace extracts later without redesign.

13. Out of Scope and Future Roadmap

The approved business case excludes API integrations, vendor-side work, licences and cloud costs. The following needs were raised and are logged for later phases:

- Weighbridge-to-XBS connector for gross (intake) and tare/exit weights (Tanzania has this connector and Uganda is building one; Kenya to adopt the same model).
- Direct XBS and Comtras APIs replacing file extracts (XBS API availability to be confirmed with the vendor).
- NCE auction data feed and automated catalogue intake.
- eDocs shipping-document generation from the shipping instruction (separate initiative).
- Sample approval tracking and wider trade risk-position digitisation.
- Tracking of re-offered auction lots (noted as low priority by Trade).

14. Assumptions, Dependencies and Risks

14.1 Assumptions

- XBS and Comtras/BRIC extracts can be produced in a consistent format at the agreed cut-off.
- Finance approves the master mapping and conversion rules before build.
- Business users are available for validation and UAT.

14.2 Dependencies

- Access to the Power Platform environment, Dataverse, SharePoint, Power BI workspace and source extracts.
- Unmanaged source of the existing solution and transfer of ownership from the previous developer's account.
- Decision on where provisional stock is held (XBS module or new solution).

14.3 Risks

Risk	Impact	Likelihood	Mitigation
XBS and Comtras positions cannot be aligned historically	High	Medium	Start from an agreed opening position at go-live; reconcile forward
Downgrade policy not applied consistently in both systems	High	Medium	Agree posting rule and owner; monthly check (EX06)
Third parties do not provide balances	Medium	High	Standard template; escalation through Logistics management
Data quality in historical explanations limits AI value	Medium	Medium	Structure explanations from go-live; AI as advisory only
Users bypass the register for franchise/ex-office	Medium	Medium	Make the register the only route to reintegration; training

15. Open Questions and Decisions Required

#	Question / decision	Owner	Needed by
1	Official cut-off time and who produces the XBS and Comtras extracts	Carol Muthama	16 Oct 2026
2	Pocket and unit rule to apply in both the app and XBS; treatment of lights	Finance, Ops	16 Oct 2026
3	Where provisional stock is held (XBS module or new solution); who maintains it	Trade, IT	16 Oct 2026
4	How the monthly Low Grade downgrade is posted in XBS and Comtras, and by whom	Trade, QC, Finance	16 Oct 2026
5	Reconciliation tolerance (e.g. ≤0.10% net) and whether measured on bags, kg or value	Finance	16 Oct 2026
6	Stuffed-not-invoiced ageing thresholds and third-party ageing limit (21 days vs other values heard)	Trade, Logistics	23 Oct 2026
7	Sign-off order and distribution list	Carol Muthama	16 Oct 2026
8	Confirm figures heard in sessions: P04 variance breakdown, basket comparison, ~USD 600k season-end loss	Finance, Trade	16 Oct 2026
9	Treatment of cancelled contracts in reconciliation	Finance, Trade	23 Oct 2026
10	Which historical reconciliations can be shared as AI context	Finance	23 Oct 2026

16. Action Items

#	Action	Owner	Due
1	Share secondary-to-primary basket mapping incl. Low Grade	Trade / QC	14 Oct 2026
2	Share grade and quality classification reference	Warehouse / QC	14 Oct 2026
3	Share P04 and recent reconciliation workbooks with explanations	Finance	14 Oct 2026
4	Share quality long/short Excel and direct purchase overview	Trade / QC	14 Oct 2026
5	Share ex-office and sweepings registers	Warehouse	14 Oct 2026
6	Share third-party confirmation template, statements and ageing report; agree the standard third-party reporting format	Logistics	14 Oct 2026
7	Confirm cut-off, unit rule, downgrade posting and provisional-stock location (Section 15)	Carol Muthama	16 Oct 2026
8	Draft master mapping and conversion rules for Finance approval	Pathways (Nicole)	16 Oct 2026
9	Draft exception catalogue, root-cause categories and routing rules for review	Pathways (Nicole, Jones)	16 Oct 2026
10	Design live count dashboard, drill-down views and category registers	Pathways (Power Apps team)	16 Oct 2026
11	Confirm XBS and Comtras extract formats and schedule; investigate XBS API availability with the vendor	Maina Iraki / Pathways (Denis)	16 Oct 2026
12	Review WIP requirements and confirm scope with Warehouse and Finance	Pathways / Warehouse / Finance	16 Oct 2026
13	Define month-end snapshot, period-locking and data-ingestion approval rules	Pathways (Denis) / Finance	16 Oct 2026
14	Review duplicate-lot and location validation rules; define part-lot tracking across locations	Pathways / Warehouse	23 Oct 2026
15	Define management and reintegration workflow for disputed, loose-bag, sweepings, ex-office and franchise stock; confirm with Quality	Pathways / Trade / QC	23 Oct 2026
16	Define stuffed-not-invoiced ageing thresholds and alert rules	Trade / Finance	23 Oct 2026
17	Design the variance explanation log and comment section for Logistics and other departments	Pathways (Power Apps team)	23 Oct 2026
18	Produce SOPs and training material for counter, warehouse, Trade, QC and Logistics users	Pathways / Operations	Before Oct count

17. Success Criteria

Measure	Baseline (P04 2026)	Target
Days from period end to signed reconciliation	33 calendar days	≤ 7 working days
Level 2 reconciliation effort	~2 weeks	≤ 2 days
Person-days of analysis per cycle	15–20	≤ 7
Count duration	2–3 days + 1–2 days re-verification	1 day confirmation
Net variance	2,824 bags to reconcile at close	≤ 0.10% (basis to confirm)
Variances with a recorded root cause	Not tracked	100%
Exceptions closed within 5 working days	Not tracked	≥ 95%
Stock categories visible (WIP, provisional, direct purchase, franchise, ex-office, sweepings, loose, disputed)	Not in system	All tracked
Third-party balances confirmed before count	2 of 5 warehouses	5 of 5
Sign-off	Paper	Electronic with timestamps

The project will also be considered successful when:

- Physical and system counts are reconciled automatically, and XBS and Comtras positions are reconciled daily.
- Variances are identified immediately, with a root-cause classification, and duplicate counting is minimised.
- Third-party balances are validated before the stock take.
- Historical stock positions are preserved, and users can trace inventory to warehouse, location, lot and grade.
- Management dashboards give real-time visibility of stock-take progress.
- Stuffed-not-invoiced variance is tracked, aged and explained, and contract roll-over linkages are traceable.
- Provisional stock is managed from contracting to physical receipt, and franchise, ex-office, loose bags and sweepings are tracked and reintegrated through a structured workflow.
- Logistics and other departments record variance explanations directly in the application, replacing follow-up calls and emails.

Annex A — Solution Build Register

This register shows what the delivery team will build and which pain point each item resolves. It is completed by the Pathways delivery team and reviewed with the business owner.

How to complete the register

- Rows are pre-filled with the items proposed in this BRD. Confirm or refine each row, add new rows at the bottom of the table, and enter the developer responsible.
- **Pain point addressed:** use the IDs in Sections 5–8 (F = Finance, W = Warehouse & Operations, T = Trade & QC, L = Logistics) and exceptions EX01–EX14.
- **Resolution achieved:** describe the business outcome in plain language. Update it at UAT with the evidence (test script or demo).
- **Req. ref:** the requirement IDs in Section 9 that the item delivers.

A.1 Power Apps

#	Task	Description	Pain point addressed	Resolution achieved	Req. ref	Developer
PA-01	Count capture enhancements	Capture bags, pocket weight and packing type; apply one conversion rule for pockets, lights and loose bags; show units, kg and 60 kg equivalents live	F6, W6, W11	Counts and reports use the same units; no false variances from lights or pockets	FR-11	
PA-02	Duplicate and multi-location warnings	Warn when a lot is already counted or sits in several locations; show where; record with a flag	W3, EX04	Duplicate counting minimised; reconciliation no longer inflated	FR-31	
PA-03	Found-elsewhere capture	Record coffee found in a different warehouse from XBS with its actual location and raise a data-correction exception	W4, EX03	Counters are not blocked; location errors corrected at source	FR-32	
PA-04	Stock category registers	Registers for franchise, ex-office, sweepings, loose bags and disputed stock; WIP, processing and direct-purchase shown separately; add, count, adjust, release into XBS	F11, W5, T9	All stock categories visible with a record; no more "symbolic" purchases	FR-21 to FR-23	
PA-05	Provisional stock pipeline	Track direct purchases from contract → warrant → delivery with quantities at each stage; include in counts	F5, T7	Direct-purchase gaps visible before they become variances	FR-20, FR-60	

#	Task	Description	Pain point addressed	Resolution achieved	Req. ref	Developer
PA-06	Master mapping maintenance	Maintain outturn → quality code → secondary → primary basket → contract, with effective dates and Finance approval	T1, F3	One shared key across app, XBS and Comtras	FR-10	
PA-07	Basket transitions and downgrade worksheet	Show processing results, secondary-basket transitions and quality results; monthly Low Grade downgrade worksheet	T4, T11, T13	Processing and downgrades traceable; Comtras aligned with physical baskets	FR-12, FR-14	
PA-08	Exception register and comment log	Exceptions with owner, root cause, due date, evidence, comments and resolution status	F4, F8, L2	Every variance has a recorded reason; no follow-up calls and emails	FR-50 to FR-53	
PA-09	Third-party confirmation tracker	Standard request, response, balance vs XBS and escalation per third-party warehouse	L1, F12, EX10	Third-party balances validated before the count	FR-65	
PA-10	Location management	Warehouse, zone, grid, location and lot; every movement linked to a unique reference	F13	Stock traceable for investigations and audits	FR-34	
PA-11	Reconciliation workspace and sign-off	Level 1 and Level 2 views with reconciling items; electronic sign-off with approver, timestamp and status	F3, F9	Reconciliation signed electronically within the SOP timeline	FR-40, FR-41, FR-70	
PA-12	Pre-count part-lot and in-transit list	List of part lots and in-transit coffee by warehouse before the count	L4, EX14	Interim variances explained before they occur	FR-33	
PA-13	Data-correction batch	Collect approved corrections into one clean-up list for XBS	L5, EX13	One clean-up instead of many single corrections	FR-56	
PA-14	Contract lineage view	Old → new contract links with residual balances	T3	Roll-over balances traceable	FR-64	
PA-15	Offline-tolerant capture	Count capture continues on weak network and syncs later	F10	No lost count time at third-party sites	NFR-05	

A.2 Power Automate

#	Task	Description	Pain point addressed	Resolution achieved	Req. ref	Developer
PF-01	Automated extract intake	Receive XBS and Comtras/BRIC extracts on schedule, validate (columns, dates, control totals) and load; reject with reason	F1, W1	No manual download/upload; one controlled extract for all teams	FR-01, FR-02, FR-05	
PF-02	Period-end snapshot and lock	Freeze the month-end position and record the extract used	F2, W12	Single version of the truth; history unchanged after processing	FR-03, FR-04	
PF-03	Daily exception run and alerts	Run exception rules daily; notify owners in Teams/Outlook; escalate when overdue	F7, T12	Issues found on the day they occur, not at month-end	FR-52, FR-54	
PF-04	Processing-complete notification	Notify QC when processing results are posted in XBS	W8	QC no longer follows up by phone	FR-55	
PF-05	Third-party requests and ageing reminders	Send confirmation requests, chase, escalate; remind before the 21-day limit	L1, L3, EX10, EX11	Balances returned on time; storage cost controlled	FR-65, FR-66	
PF-06	Stuffed-not-invoiced ageing alerts	Alert when stuffed coffee passes the invoicing threshold	T6, EX07	Working capital released sooner	FR-43	
PF-07	Contract amendment-limit alert	Alert Trade as a contract approaches 100 amendments; record lineage	T3, EX12	Roll-overs planned, not discovered	FR-64	
PF-08	Catalogue duplicate check	Compare direct and auction catalogues on upload and alert duplicates	T8, EX09	No double purchase of the same coffee	FR-61	
PF-09	Allocation report generation	Produce the allocation report in BRIC/internal format from XBS stuffing data and distribute it	T2	No manual reformatting per shipment	FR-63	
PF-10	Sign-off workflow and pack distribution	Route the reconciliation for approval in order; send the sign-off pack to the distribution list	F9	Electronic sign-off with audit trail	FR-70, FR-71	
PF-11	Correction batch to XBS owner	Send approved corrections as one batch and confirm posting	L5, EX13	Corrections posted in one clean-up	FR-56	
PF-12	Warehouse validation at upload	Check warehouse and quantity against master mapping, warrant and movement instruction	W7	Wrong-warehouse uploads caught before the count	FR-62	

A.3 Dashboards (Power BI)

#	Dashboard	KPIs	Pain point addressed	Resolution achieved	Insight	Refresh frequency	Source
DB-01	Live count (Physical vs System vs Variance)	% locations counted; counted vs not; found-not-in-XBS; in-XBS-not-found; wrong warehouse; duplicates	W2, F7	Count-day gaps visible live	Where to recount or investigate now	Every 15 min on count day	Stock Take app; XBS snapshot
DB-02	Level 1 reconciliation	Variance in bags, kg and 60 kg eq. by outturn, location, grade; count accuracy %	F3, F6	Level 1 explained in one view	Which outturns and locations drive the variance	Each count	Stock Take app; XBS snapshot
DB-03	Three-way reconciliation	XBS vs Comtras vs quality long/short by basket and contract; reconciling items; unexplained residual	T5, T12, F3	Level 2 in days, not weeks	Which reconciling items explain the difference	Daily; month-end certified	XBS; Comtras 526/537; quality long/short
DB-04	Exceptions and root causes	Open, overdue, by owner and root cause; % with root cause; % closed within 5 days	F4, F7, L2	Every variance owned and explained	Recurring error families and owners	Daily	Exception register
DB-05	Stock categories	Balances and movements for provisional, franchise, ex-office, sweepings, loose, disputed, WIP, processing	F11, W5, T9	All categories in one view	Growing categories needing action	Weekly	Category registers; XBS
DB-06	Stuffed-not-invoiced ageing	Bags and value by contract; days since stuffing; past threshold	T6	SNI tracked and aged	Contracts holding up invoicing	Daily	XBS stuffing; Comtras/BRIC
DB-07	Third-party position and ageing	Balance vs XBS; confirmation status; days held; storage exposure	L1, L3, F12	Third-party stock confirmed and controlled	Warehouses at risk of exceeding limits	Weekly	Third-party tracker; XBS
DB-08	Shorts in transit and shipped in month	Contracted/dispatched vs received weight; shipments by contract and date	L4, F3	Movement variances explained	Suppliers and routes with recurring shorts	Monthly	XBS; Comtras; Logistics records
DB-09	Basket transitions and downgrades	Processing in/out by basket; downgrade to Low Grade; posted vs not posted	T4, T11	Comtras baskets match physical stock	Processing loss by month and basket	Monthly	XBS processing; Comtras
DB-10	Governance KPI scoreboard	Days to sign-off; variance %; exceptions closed on time; same-day posting	F9, F2	SOP timeline tracked	Whether the 7-day target is being met	Monthly	All of the above

#	Dashboard	KPIs	Pain point addressed	Resolution achieved	Insight	Refresh frequency	Source

A.4 Copilot Studio

#	Task	Description	Pain point addressed	Resolution achieved	Req. ref	Developer
CP-01	Variance explanation	Answer "why is this outturn/basket different?" using the reconciliation and recorded explanations	F4, F8	Faster understanding of variances	FR-80	
CP-02	Root cause and next step	Suggest a likely root cause and next action for an exception; user confirms	F4, EX01–EX14	Exceptions classified consistently	FR-80	
CP-03	Stock lookup	Answer where an outturn is, its category, basket and open exceptions	F13, W4	Fewer calls between departments	FR-80	
CP-04	Count progress	Answer count-day questions: what is left, what is not in XBS	W2	Supervisors see progress without opening reports	FR-80	
CP-05	Comment summary	Summarise exception comments for the post-stock-take review and sign-off pack	L2, F9	Review meetings shorter and better prepared	FR-80	
CP-06	Rules and SOP questions	Answer questions on unit rules, categories and the stock-take SOP	W9, F6	Consistent practice across counters and teams	FR-90	
CP-07	Historical explanations knowledge	Load recent reconciliation explanations in a structured form as context	F8	Assistant answers from real history	FR-81	

All Copilot outputs are advisory and subject to human review (FR-80).